

HEXAGON MULTI-SENSOR ENVIRONMENTAL DETECTOR

Technical Engineering Specification v1.0

1. EXECUTIVE OVERVIEW

Product Name: HexaGuard-6 Environmental Safety Detector

Form Factor: Regular hexagon (diameter: 150mm, height: 85mm)

Detection Capabilities: 6-gas + mold VOC environmental monitoring system

Primary Markets: Residential, light commercial, industrial facilities

2. PHYSICAL DESIGN SPECIFICATIONS

2.1 Hexagonal Housing

- Outer Dimensions:** 150mm hexagon (flat-to-flat: 130mm)
- Height:** 85mm
- Material:** ABS polymer + aluminum thermal heatsink base
- Wall Thickness:** 3.2mm (ABS), 2mm (aluminum)
- Weight:** 420g (fully assembled)
- IP Rating:** IP54 (dust and splash resistant)
- Operating Temperature:** -10°C to +50°C
- Operating Humidity:** 15% to 95% RH (non-condensing)

2.2 Hexagonal Layout & Sensor Positioning

Each of six hexagonal faces houses a primary sensing module:

Face 1 (Front/Main): Photoelectric Smoke Detector + Primary LED Array

Face 2: Carbon Monoxide Sensor + Display Window

Face 3: NDIR CO2 Sensor Module

Face 4: Hydrogen Sulfide + Flammable Gas Combo Sensor

Face 5: VOC/Mold Detection Array

Face 6: Air Circulation & Calibration Port

2.3 Mounting

- Wall-mount bracket (standard 90mm pitch)
 - Alternative ceiling mount adapter (included)
 - Quick-release connector with 24-month battery backup
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3. SENSOR SPECIFICATIONS

3.1 SMOKE DETECTION (Photoelectric)

Technology: Dual-wavelength photoelectric sensor

Specifications:

- **Sensor Model:** Honeywell FDS161-AEX (or equivalent)
- **Detection Method:** Light scattering at 850nm + 950nm wavelengths
- **Sensitivity:** NFPA 72 compliant ($\geq 0.5\text{dB/m}$ obscuration per foot)
- **Response Time:** <15 seconds to standard test fire (TT5 smoke)
- **False Alarm Rate:** <1 per 1000 detector-years
- **Dual Optical Paths:** Redundant detection circuits
- **Angular Response:** 360° detection zone within 3-meter radius
- **Self-Diagnostics:** Weekly optical path verification

Advantages over ionization: Better at detecting smoldering fires, reduced alpha particle chamber maintenance

3.2 CARBON MONOXIDE (CO) DETECTION

Technology: Dual-electrode electrochemical cell

Specifications:

- **Sensor Model:** City Technology Ltd. 4CO (or equivalent)
- **Detection Range:** 0–1000 ppm
- **Alarm Thresholds (UL 2034 & EN 50291):**
 - 70 ppm → Alarm within 60 minutes
 - 150 ppm → Alarm within 10–40 minutes
 - 400 ppm → Alarm within 3–10 minutes
 - 1200 ppm → Alarm within 2–3 minutes
- **Response Time:** <40 seconds from gas exposure to sensor
- **Cross-Sensitivity:** <3% to common gases (H₂S, SO₂)
- **Operating Life:** 7–10 years
- **Warm-up Time:** 60 seconds to full sensitivity

Electrochemical Cell Details:

- Dual working electrodes + reference electrode
- Silver/silver chloride reference system
- Electrolyte: Sulfuric acid (H₂SO₄) based
- Membrane: Teflon PTFE diffusion barrier
- Calibration Interval:** Annual (recommended)

3.3 CARBON DIOXIDE (CO₂) DETECTION

Technology: Non-Dispersive Infrared (NDIR) Sensor

Specifications:

- **Sensor Model:** SenseAir K30 (or equivalent)
- **Detection Range:** 0–5000 ppm (expandable to 10,000 ppm)
- **Accuracy:** ±50 ppm + 5% of reading
- **Resolution:** 1 ppm
- **Response Time (T₉₀):** <20 seconds
- **Measurement Interval:** Configurable (default: 2 seconds)
- **Dual-Beam Optics:** Filters at 4.26 µm wavelength

- **Temperature Compensation:** Automatic internal sensor
- **Calibration Method:** Automatic baseline correction (ABC logic)
 - Assumes lowest reading in 24-hour period = 400 ppm outdoor level
 - Manual override available via service port

Optical Path Specifications:

- Infrared LED source: 3.3V, 250mW peak
 - Two optical filters (4.26 μm , 3.9 μm reference)
 - Thermopile detector with integrated electronics
 - Optical path length: 10mm internal chamber
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3.4 HYDROGEN SULFIDE (H₂S) DETECTION

Technology: Electrochemical 3-electrode cell

Specifications:

- **Sensor Model:** Alphasense OX-H₂S-100 (or equivalent)
- **Detection Range:** 0–100 ppm
- **Alarm Threshold:** 10 ppm (OSHA PEL for 8-hour exposure)
- **Response Time (T₉₀):** <20 seconds
- **Accuracy:** ± 5 ppm or $\pm 10\%$ of reading
- **Cross-Sensitivity:**
 - SO₂: <2% (minimal interference)
 - CO: Negligible
 - Chlorine: <1%
- **Long-term Stability:** <5% drift over 6 months
- **Expected Lifespan:** 3–5 years
- **Temperature Range:** -20°C to +50°C (sensor specific)

Cell Configuration:

- Three-electrode potentiostat system
- Working electrode: Gold with selective H₂S coating

- Counter & reference electrodes: Silver/AgCl
 - Electrolyte: Polymer gel matrix
 - Membrane selectivity: Prevents large molecules (H₂, N₂)
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3.5 FLAMMABLE GAS VAPOR DETECTION

Technology: Catalytic Bead + PID Hybrid

Primary Sensor (Catalytic Bead):

- **Sensor Model:** Dräger Polytron 8750 cartridge or equivalent
- **Detection Range:** 0–100% LEL (Lower Explosive Limit)
- **Typical Target Gases:** Methane, propane, natural gas, benzene vapors
- **Response Time (T₉₀):** <15 seconds
- **Repeatability:** ±2% full scale
- **Zero Drift:** <2% per year

Catalytic Bead Mechanism:

- Two matched ceramic beads coated with platinum catalyst
- Reference bead: Inert coating (baseline)
- Active bead: Catalytic combustion of flammable vapors
- Operating Temperature: Electrically heated to 500°C
- Resistance change measured via Wheatstone bridge
- Output: 4–20mA current loop

Secondary Sensor (PID - Photoionization Detector):

- **Sensor Model:** BMT 932 (or equivalent compact PID)
- **Detection Range:** 0–100 ppm
- **Ionization Energy:** 10.6 eV UV lamp
- **Response Time:** <5 seconds
- **Cross-sensitivity:** Detects volatile organics, solvents
- **Redundancy:** Confirms catalytic bead readings via different mechanism

Hybrid Logic:

- Both sensors operate simultaneously
- Agreement correlation: $\pm 15\%$ variance acceptable
- Mismatch triggers advanced diagnostics alert
- PID sensitivity to aromatics complements bead catalytic selectivity

3.6 BLACK MOLD & MYCOTOXIN DETECTION (VOC Array)

Technology: Electrochemical VOC Sensors + Machine Learning

Primary Sensors:

- **Sensor 1 - Formaldehyde:** Alphasense OX-FORM-100
 - Range: 0–100 ppm
 - Typical mold byproduct detection
- **Sensor 2 - General VOCs:** Alphasense OX-A2
 - Detects organic compounds including musty odors
 - Range: 0–500 ppm equivalent
- **Sensor 3 - Mycotoxin Markers:** Electrochemical sensor for ethyl benzene/toluene
 - Detects secondary metabolites from *Aspergillus* & *Stachybotrys* species
 - Range: 0–50 ppm

Detection Algorithm:

- **Machine Learning Profile Matching:** Trained on 500+ mold species VOC signatures
- **Sensor Fusion:** Combines all three VOC channels
- **Baseline Learning:** Self-calibrates to building's normal VOC profile
- **Distinctive Pattern Recognition:**
 - *Aspergillus niger*: High acetone + ethanol signature
 - *Stachybotrys chartarum*: Geosmin-like compound + alcohols
 - *Penicillium*: Characteristic 2-methylisoborneol (2-MIB)

Confidence Scoring:

- 0–30%: Possible mold presence (advisory)
- 30–70%: Probable mold contamination (warning)
- 70–100%: High confidence mold growth (alert)

Sensor Specifications:

- Response time: 30–60 seconds (slower than gas sensors due to algorithm processing)
 - Warm-up time: 3–5 minutes to full sensitivity
 - Lifespan: 2–3 years for VOC sensors
 - Cross-interference correction: Automatic compensation for cooking, cleaning products
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4. ELECTRONIC CONTROL ARCHITECTURE

4.1 Central Processor

Microcontroller: ARM Cortex-M4F (STM32L476RG)

- 80 MHz clock speed
- 1 MB Flash, 128 KB SRAM
- Ultra-low power mode (<50µA standby)
- Dual 12-bit ADC, 5 MSPS
- Built-in RTC with backup battery

4.2 Sensor Interface Electronics

Analog Signal Conditioning:

- 6-channel precision instrumentation amplifier (INA128)
- Individual programmable gain (1–100x) per channel
- 24-bit delta-sigma ADC (ADS1256) for high-resolution acquisition
- Low-noise design: <0.5 mV RMS input-referred noise

Electrochemical Sensor Potentiostat:

- Precision voltage regulation: $\pm 10\text{mV}$ stability
- Current measurement: 0–200nA to 1mA range
- Temperature compensation: Built-in thermistor on sensor boards
- Digital potentiometer control for bias voltage adjustment

NDIR CO₂ Sensor Interface:

- Dedicated 3.3V regulated power supply (250mA capacity)
- I2C communication protocol (400 kHz)
- Warm-up sequence: Gradual power ramp over 30 seconds
- Optical stabilization: Temperature monitoring to thermopile output

4.3 Power Management

Primary Power: 2 × AA alkaline batteries (3V nominal)

- Expected lifespan: 5–7 years (standard operation)
- Total capacity: 5000 mAh (2500 mAh per cell)

Battery Chemistry Consideration:

- Alkaline (standard, cost-effective)
- Alternative: Energizer Max (marginal improvement for high-drain sensors)
- NOT recommended: Zinc-carbon (insufficient current capability)

Secondary Backup: CR123A lithium cell (3V)

- 24-hour emergency operation minimum
- Auto-selects if primary batteries fail
- Independent low-battery detection

Power Budget (Operating):

Component	Current (mA)	Duty Cycle	Average
Microcontroller	15	100%	15 mA
Smoke Sensor (LED)	8	10%	0.8 mA
CO Electrochemical	2	100%	2 mA
CO ₂ NDIR (LED)	80	2%	1.6 mA
H ₂ S Sensor	3	100%	3 mA

Flammable Gas (Catalytic)	250	5%	12.5 mA
VOC Sensors (3×)	5	100%	5 mA
LEDs & Buzzer	50	1%	0.5 mA
Wireless (optional)	20	2%	0.4 mA
Total Average	—	—	**41.2 mA**

Operating Duration: 5–7 years continuous (alkaline batteries)

5. SIGNAL PROCESSING & ALGORITHMS

5.1 Real-Time Data Fusion

Sampling Rate: 10 Hz across all sensors

Processing Pipeline:

1. **Analog Acquisition:** Simultaneous sampling of all 6 sensor channels
2. **Baseline Normalization:** Zero-offset correction per sensor
3. **Time-Weighted Filtering:**
 - Smoke: 5-point moving average (500ms window)
 - CO: 8-point moving average (800ms window)
 - CO₂: 4-point moving average (400ms window)
 - H₂S: 6-point moving average (600ms window)
 - Flammable: 3-point moving average (300ms window)
 - VOC Array: 20-point moving average (2000ms window)
4. **Cross-Correlation Detection:** Check sensor agreement
5. **Alarm Decision Logic:** Multi-threshold evaluation

5.2 Alarm Logic & Thresholds

Priority Tier 1 (Immediate Evacuation Alarm):

- Smoke: >0.5 dB/m obscuration for 3 consecutive seconds
- CO: >400 ppm for >10 seconds
- Flammable: >25% LEL sustained

- **Action:** Siren at 85 dB(A), LED flash (1 Hz red), wireless alert

Priority Tier 2 (Warning - Investigate):

- CO: 70–150 ppm sustained for 5 minutes
- H₂S: >10 ppm for 60 seconds
- CO₂: >1500 ppm (ventilation issue indicator)
- Mold Confidence: 60–70%
- **Action:** LED orange flash (0.5 Hz), buzzer at 70 dB(A) (intermittent)

Priority Tier 3 (Advisory - Monitor):

- CO₂: 1000–1500 ppm (CO₂ buildup warning)
- Mold Confidence: 30–60%
- **Action:** LED yellow steady, no sound alert (notifications only)

5.3 Self-Diagnostics (Weekly)

Test Sequence:

- Verify all sensor responses to internal test pulses
- Check battery voltage (alert if <2.5V)
- Verify optical path cleanliness (photoelectric sensor)
- Confirm wireless connectivity (if applicable)
- Temperature & humidity compensation verification
- **Reporting:** Transmit diagnostic data; local LED indicates "OK" (green 2-second pulse)

6. COMMUNICATION & ALERTS

6.1 Local Alerts

LED Array (RGB LEDs - 3mm, 20mA max):

- Green (charging): Standby/normal operation
- Orange (flash): Warning level alarm
- Red (rapid flash): Emergency alarm

- Amber (breathing): Low battery warning

Acoustic Alert:

- Alarm siren: 85 dB(A) at 1 meter
- Warning tone: 70 dB(A) intermittent
- Frequency: 3400 Hz (optimal human hearing range)

6.2 Wireless Connectivity (Optional Module)

Technology: 802.15.4 (Zigbee-compatible) or WiFi (802.11 b/g/n)

Features:

- Real-time sensor data streaming (1 Hz)
 - Cloud logging with 30-day history
 - Mobile app notifications
 - Range: 30–50 meters (residential), extendable via mesh network
 - Power draw: 20 mA average (periodic transmission)
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7. CALIBRATION & MAINTENANCE**7.1 Factory Calibration****Electrochemical Sensors:**

- Zero calibration: Pure nitrogen exposure at manufacturing
- Span calibration: Certified gas mixture (CO: 100 ppm, H₂S: 20 ppm)
- Temperature compensation: -10°C, +20°C, +50°C reference points

Smoke Detector:

- TT5 standard test fire
- Angular response verification (all 6 hexagon faces)

CO₂ NDIR:

- Zero calibration: Pure CO₂-free air (400 ppm reference)
- Span: 5000 ppm calibration gas

7.2 In-Service Maintenance

Annual Service (Recommended):

- Visual inspection for dust accumulation
- Battery voltage check
- CO sensor replacement (every 7 years)
- Electrochemical cell replacement (every 3–5 years)
- VOC sensor array replacement (every 2–3 years)

5-Year Component Replacement:

- All sensor cartridges
- Electrolyte replenishment in electrochemical cells
- Optical lens cleaning (smoke detector)

Professional Calibration Interval: Every 2 years (recommended for commercial installations)

8. CERTIFICATIONS & COMPLIANCE

8.1 Safety Standards

- **Smoke Detection:** NFPA 72 (National Fire Alarm Code), UL 217, EN 14604
- **CO Detection:** UL 2034, EN 50291
- **Multi-gas Industrial:** ATEX Category 3 (Group II, Category 3 for hazardous areas)
- **EMC:** FCC Part 15, CE Mark (EN 61000-6-2 industrial immunity)
- **RoHS Compliance:** Directive 2011/65/EU

8.2 Material Safety

- **RoHS 2 Compliant:** Lead-free solder, no mercury, no cadmium
- **Halogen-Free:** PCB substrate per IEC 61249-2-21
- **Electrochemical Cell Electrolyte:** MSDS available; non-toxic gel in sealed chambers

9. TECHNICAL PERFORMANCE SUMMARY

Parameter	Value	Notes
Overall Dimensions	150mm hexagon × 85mm	Compact wall/ceiling mount
Weight	420g	Aluminum heatsink included
Battery Life	5–7 years	Alkaline AA cells
Sensor Response Time	5–60 seconds	Varies by sensor type
Operating Range	6 gases + VOC	All simultaneous
Accuracy	±5–10% typical	Per individual sensor specs
Alarm Latency	<2 seconds	From detection to alert
False Alarm Rate	<0.1% per year	Validated by dual-channel design
IP Rating	IP54	Dust & splash resistant
Operating Temperature	-10°C to +50°C	Full sensor capability maintained
Wireless Range	30–50m (optional)	Zigbee/WiFi capable
Self-Test Interval	Weekly	Automatic diagnostics

10. DESIGN RATIONALE & TRADE-OFFS

10.1 Hexagonal Form Factor

Advantages:

- 360° sensor exposure on 6 faces (vs. 4 for square, poor coverage for circle)
- Elegant visual design
- Increased surface area for heat dissipation
- Stable on any flat surface during testing

Disadvantages:

- Slightly higher manufacturing cost (custom injection molds)
- Unique bracket design vs. standard circular mounts

10.2 Dual-Wavelength Photoelectric (vs. Ionization)

Choice: Photoelectric for primary smoke detection

- Superior response to smoldering fires (better detection of flaming fires too)
- No radioactive alpha source (Am-241)
- Lower ongoing maintenance
- Dual optical paths reduce false alarms from dust

10.3 Electrochemical Sensors (vs. MOSFET/Metal Oxide)

Choice: Electrochemical for CO, H₂S, VOC

- Lower operating temperature (no heat-driven drift)
- Excellent selectivity (gas-specific membranes)
- Linear response across operational range
- Lower power consumption
- Trade-off: Shorter lifespan (3–7 years vs. 10+ for metal-oxide)

10.4 Machine Learning for Mold Detection

Rationale:

- Single VOC sensor cannot reliably identify mold species
- Fusion of 3 VOC channels + pattern recognition achieves 70–85% accuracy

- Alternative: PCR or microscopy (requires professional sampling, not real-time)
 - Trade-off: Requires 7–14 day learning period after installation
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11. TYPICAL DEPLOYMENT SCENARIO

Residential Installation:

- Primary detector: Hallway (central location)
- Secondary detectors: Master bedroom, kitchen periphery (≥ 3 meters from cooking)
- Basement: Radon + H₂S monitoring (water damage risk)
- Nursery: Low-alarm mode (sensitive CO detection)

Wireless Mesh Network:

- Up to 15 hexagon units in single dwelling
 - Hub transmits to cloud/mobile app
 - Battery status monitored centrally
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12. FUTURE ENHANCEMENT PATHS

1. **Radon Detection:** Add alpha-track scintillation chamber (20-day averaging)
 2. **Volatile Anaesthetic Leaks:** Extend PID range to 50 ppm (medical facilities)
 3. **Biological Hazard Detection:** E-nose array for airborne pathogenic markers
 4. **Humidity-Compensated Mold Inference:** Integrate DHT22 sensor ($\pm 1\%$ RH accuracy)
 5. **Particulate Matter (PM_{2.5}/PM₁₀):** Laser particle counter module
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APPENDIX A: SENSOR DATASHEETS & PART NUMBERS

Component	Manufacturer	Model	QTY	Notes
Smoke Sensor	Honeywell	FDS161-AEX	1	Photoelectric dual-wavelength
CO Sensor	City Tech	4CO	1	Electrochemical 0–1000 ppm
CO ₂ Sensor	SenseAir	K30	1	NDIR 0–5000 ppm
H ₂ S Sensor	Alphasense	OX-H2S-100	1	Electrochemical 0–100 ppm
Formaldehyde	Alphasense	OX-FORM-100	1	VOC detection
General VOC	Alphasense	OX-A2	1	VOC array
Ethyl Benzene	Alphasense	OX-EB-50	1	Mold marker detection
Flammable Gas	Dräger	Polytron 8750	1	Catalytic bead 0–100% LEL
PID Complement	BMT	932	1	Backup flammable detection
Microcontroller	STM	STM32L476RG	1	Cortex-M4F processor
ADC	TI	ADS1256	1	24-bit high-resolution
Op-Amp (×2)	TI	OPA2333	2	Precision instrumentation

Document Version: 1.0

Last Revised: July 2026

Classification: Engineering Specification (Public Domain)

Prepared for: Advanced Safety Systems Design

Authorization: Chief Engineering Officer